| Question | | | Answer | Marks | AO | Guidance | |
|---|---|---|---|---|---|---|---|
| **11** | | | Let $u = 2x + k$ $2\mathrm{d}x = \mathrm{d}u$ | **M1** | **2.1** | Substituting $u = 2x + k$<br>Allow for a different substitution giving an integral in $u$ only | Allow for $\int \frac{a}{u^2}\mathrm{d}u$ for any constant seen |
| | | | $\int \frac{2}{(2x+k)^2}\mathrm{d}x = \int \frac{1}{u^2}\mathrm{d}u$ | **A1** | **2.1** | Correct integrand in terms of $u$<br>Ignore limits | |
| | | | $= -\frac{1}{u}[+c]$ | **A1** | **2.1** | correct indefinite integral<br>constant need not be seen | |
| | | | $\int_k^{2k} \frac{2}{(2x+k)^2}\,\mathrm{d}x = \int_{3k}^{5k}\left(\frac{1}{u^2}\right)\mathrm{d}u = -\frac{1}{5k} + \frac{1}{3k}$ | **M1** | **2.1** | substituting correct new limits into their integrated expression, or substituting in terms of $x$ and using original limits | |
| | | | Alternatively, by inspection | | | | |
| | | | $\int_k^{2k} \frac{2}{(2x+k)^2}\,\mathrm{d}x = \left[-(2x+k)^{-1}\right]_k^{2k}$ | **M1** | | Integrating by inspection to obtain any multiple of $(2x+k)^{-1}$ | |
| | | | | **A2** | | Fully correct indefinite integral – need not be simplified. | |
| | | | $-\frac{1}{5k} + \frac{1}{3k}$ | **M1** | | substituting limits into their integrated expression | |
| | | | $= \frac{2}{15k}$ | **A1** | **2.1** | Allow $\left(-\frac{1}{5} + \frac{1}{3}\right)\frac{1}{k}$ seen | |
| | | | This is inversely proportional to $k$<br>[with constant of proportionality $\frac{2}{15}$ ] | **E1** | **2.2a** | FT their definite integral<br>Must use phrase "inversely proportional" to $k$ or indicates $\propto \frac{1}{k}$ | Allow if $\frac{a}{k}$ required at the start of the argument |
| | | | | **[6]** | | | |